Hadi Reisizadeh

+1-612-245-0654 hadi.reisizadeh@gmail.com Website Google Scholar LinkedIn

Summary

Research Scientist specializing in **LLM safety, optimization, and trustworthy AI**. Exposed critical vulnerabilities in LLM unlearning: sensitive knowledge resurfaces under probabilistic decoding, and suppressed tool-use capabilities are recovered by ordinary **agent harnesses**. Introduced **leak@**k to quantify worst-case knowledge leakage and **RULE**, an iterative training framework achieving near-zero leakage on standard benchmarks. Developed **BLUR**, a bi-level optimization approach delivering up to **19**% stronger unlearning while preserving utility, and **HARU**, a harness-aware unlearning method cutting harness-guided recovery from **80**% to **1**.**5**%.

Technical Skills

Core Expertise	LLMs, optimization, machine unlearning, efficient training, and safe AI
Programming Languages	Python, SQL, MATLAB, R, C/C++
Machine Learning	PyTorch, HuggingFace Transformers, TensorFlow, Scikit-learn
LLM Training	DeepSpeed, HuggingFace Accelerate

Work Experience

University of Minnesota, Minneapolis, MN

Postdoctoral Associate, Optimization for AI Lab — 09/2024 – Present

Research Assistant, Information Processing Group — 09/2017 – 09/2024

- **LLM Unlearning and Model Safety.**
  - Explored the fundamental limits of unlearning in LLMs, showing that sensitive knowledge can resurface under probabilistic decoding. Introduced **leak@**k, a meta-metric to quantify worst-case knowledge leakage across k generations, and **RULE**, an iterative training framework achieving near-zero leakage on TOFU and outperforming SOTA methods on MUSE-News [paper], [code].
  - Proposed a **bi-level optimization framework (BLUR)** for LLM unlearning that prioritizes the forget objective over the retain loss, achieving up to **19**% higher unlearning efficiency and **11**.**7**% higher model utility on MUSE-News with the **LLaMA-2-7B** model, and **16**.**6**% higher efficiency on WMDP with equivalent utility using the **Zephyr-7B-Beta** model [paper], [code].
  - Showed that **agent harnesses** (reasoning guidance, schema repair) can recover tool-use capabilities suppressed by unlearning without modifying model weights. Introduced the **Harness Recovery Rate (HRR)** and proposed **HARU**, cutting HRR from **32**.**8**% to **4**.**5**% (ToolAlpaca) and **80**.**0**% to **1**.**5**% (BFCL) while preserving retained-tool utility.
- **Efficient Reasoning Condensation.** Proposed an **edge-preserving condensation framework** for chain-of-thought reasoning training that retains only the initial and final reasoning stages, enabling supervised models to achieve **lossless reasoning accuracy** while reducing training time by over **34**% across multiple **LLM families (Qwen, LLaMA)** and reasoning benchmarks [paper], [code].
- **Distributed Training.** Designed a **two-time-scale optimization algorithm** for communication-efficient distributed training over noisy, time-varying networks, achieving **provable convergence guarantees** in both convex and non-convex settings [paper 1], [paper 2], [paper 3].

Seagate Technology, Shakopee, MN — 05/2023 – 04/2024

Research Intern, Data Trust Research Team

- **Differentially Private Distributed Training.** Developed an **accelerated privacy-preserving distributed training method**, achieving faster convergence without reducing privacy guarantees.

Education

Ph.D. & M.Sc. in Electrical and Computer Engineering (*Minor in Computer Science*)

University of Minnesota, Twin Cities, Minneapolis, MN — 09/2017 – 09/2024

Selected Publications

- **H. Reisizadeh**, J. Ruan, Y. Chen, S. Pal, S. Liu, and M. Hong, *"Leak@*k*: Unlearning Does Not Make LLMs Forget Under Probabilistic Decoding,"* ICML, 2026.
- **H. Reisizadeh**, J. Jia, Z. Bu, B. Vinzamuri, A. Ramakrishna, K. Chang, V. Cevher, S. Liu, and M. Hong, *"BLUR: A Bi-Level Optimization Approach for LLM Unlearning,"* EACL, 2026.
- **H. Reisizadeh**, J. Ruan, S. Liu, and M. Hong, *"Do LLMs Really Forget? Hidden-State Leakage in Model Unlearning and How to Fix it,"* Submitted, 2026.
- J. Ruan, **H. Reisizadeh**, S. Liu, and M. Hong, *"When Unlearning Meets the Harness: Helpful Assistance, Harmful Recovery,"* Submitted, 2026.
- J. Jia, **H. Reisizadeh**, C. Fan, N. Baracaldo, M. Hong, and S. Liu, *"EPiC: Towards Lossless Speedup for Reasoning Training through Edge-Preserving CoT Condensation,"* Submitted, 2026.